

Introduction

Clustering Analysis

The purpose of this clustering analysis was to group customers with similar characteristics. For this analysis, I used the tenure and MonthlyCharges columns from the cleaned customer dataset. Tenure shows how long a customer has been with the company, while MonthlyCharges shows how much the customer pays each month. I used K-Means clustering to divide the customers into different groups. The aim was to understand the different customer segments and compare their churn behaviour.

2. Features Used and Data Preparation

For the clustering analysis, I used tenure and MonthlyCharges as the main features. These features were selected based on the project requirements. Tenure represents how long a customer has stayed with the company, and MonthlyCharges represents the amount the customer pays each month.

Before starting the clustering, I checked both features for missing values. There were no missing values in either tenure or MonthlyCharges. The cleaned dataset contained 7,043 customer records.

I then used StandardScaler to standardise the two features. This was important because tenure and MonthlyCharges have different measurement ranges. K-Means uses distance to group similar customers, so scaling helped make sure that both features had a fair influence on the clustering.

3. Selecting the Optimal Number of Clusters

To decide the number of clusters, I used the Elbow Method. I tested different numbers of clusters from $K = 1$ to $K = 10$. From the elbow graph, the decrease in inertia became smaller after $K = 4$, which suggested that four clusters were a suitable choice.

I also used the Silhouette Score to support this decision. I tested K values from 2 to 6, and $K = 4$ had the highest Silhouette Score of 0.476. Based on both results, I selected four clusters for the final K-Means model.

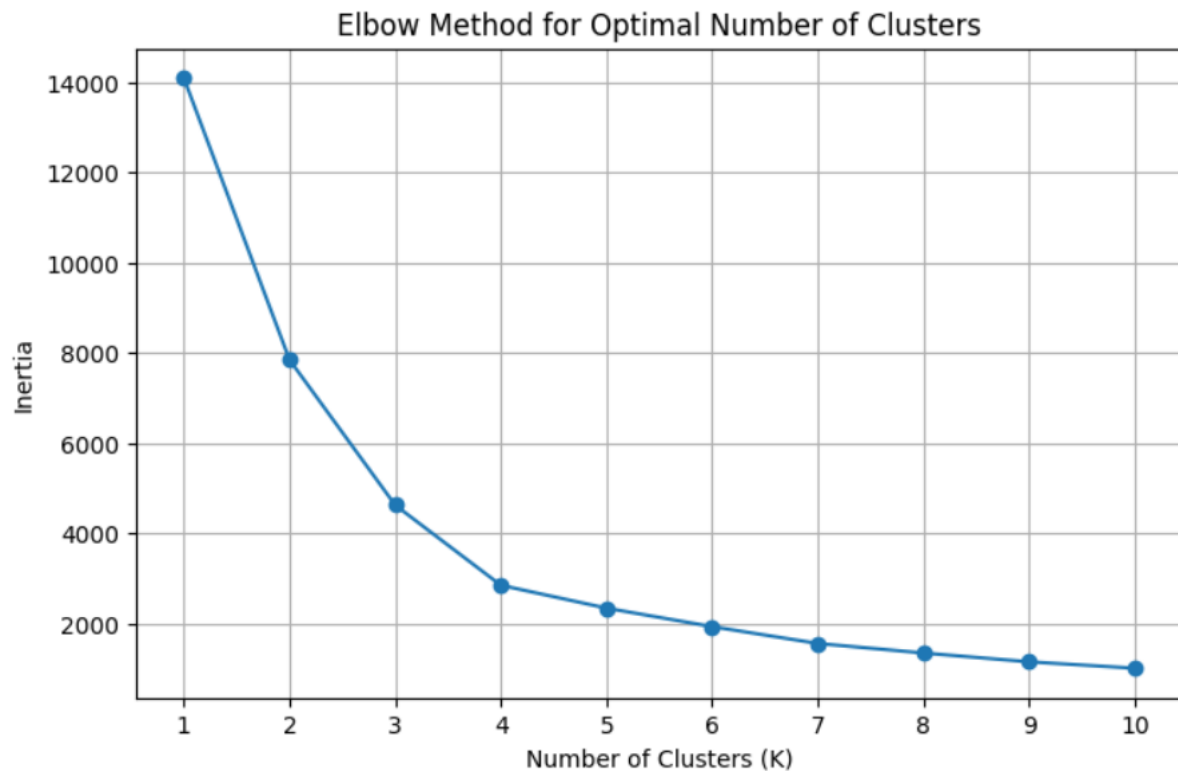


Figure 1: Elbow Method for selecting the optimal number of clusters

Number of Clusters (K)	Silhouette Score
2	0.411
3	0.446
4	0.476
5	0.430
6	0.415

4. K-Means Clustering Results

After selecting $K = 4$, I trained the final K-Means clustering model using the scaled tenure and MonthlyCharges data. The model assigned all 7,043 customers into four clusters. I then looked at the average tenure and monthly charges in each cluster to understand the characteristics of each customer group.

Based on these characteristics, I gave each cluster a simple and meaningful segment name.

Cluster	Customers	Average Tenure (Months)	Average Monthly Charges (\$)	Customer Segment
0	2,202	14.81	81.23	Newer – High Charges
1	1,733	10.48	32.49	Newer – Low Charges
2	1,948	58.76	93.08	Long-term – High Charges
3	1,160	54.10	34.15	Long-term – Low Charges

Cluster 0 represents newer customers with relatively high monthly charges, while Cluster 1 represents newer customers with lower monthly charges. Cluster 2 contains long-term customers with high monthly charges, and Cluster 3 contains long-term customers with lower monthly charges. These labels make the four customer groups easier to understand and compare.

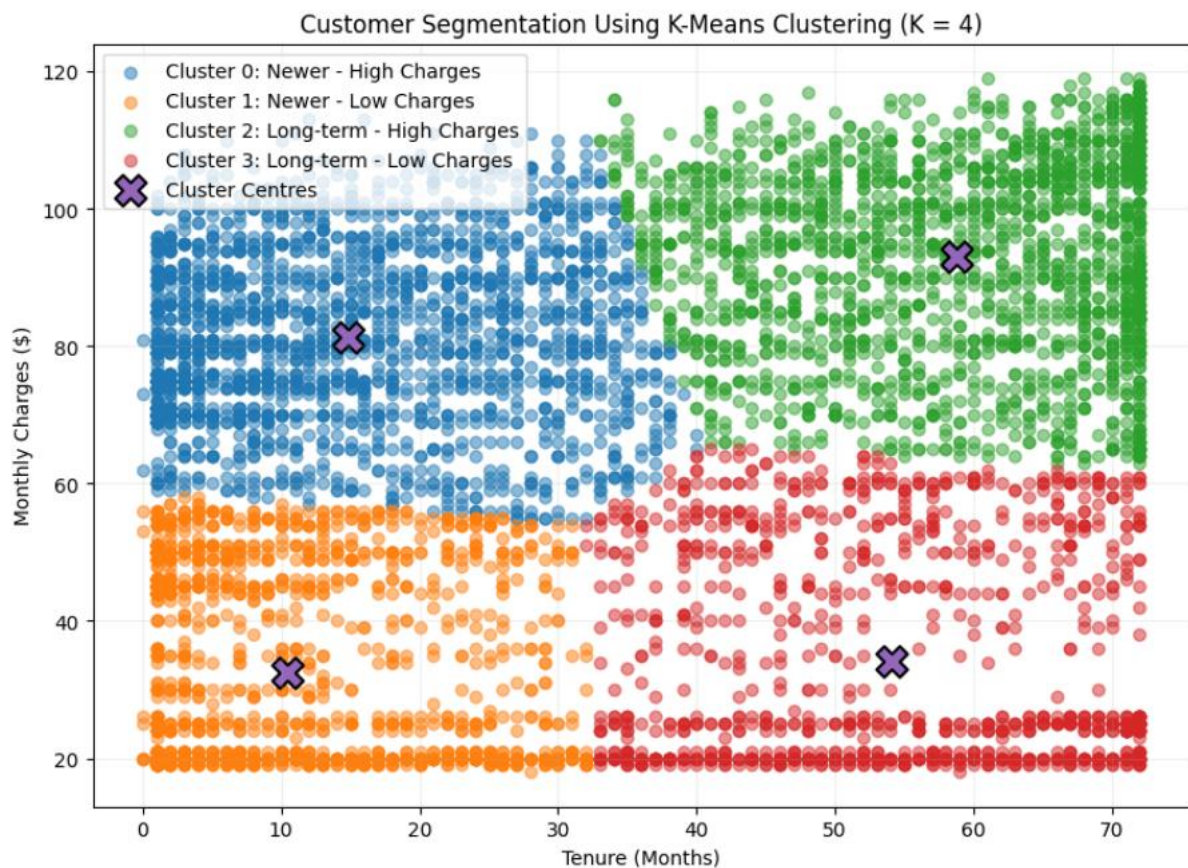


Figure 2: Customer segmentation using K-Means clustering (K = 4)

5. Churn Analysis by Customer Segment

After creating the four customer segments, I compared the churn rate for each cluster. Churn was not used to create the clusters. It was only used after the clustering process to understand how churn differed between the customer groups.

The results showed that Cluster 0 (Newer – High Charges) had the highest churn rate at 49.18%. Cluster 1 (Newer – Low Charges) had a churn rate of 24.47%, while Cluster 2 (Long-term – High Charges) had a lower churn rate of 15.71%. Cluster 3 (Long-term – Low Charges) had the lowest churn rate at only 4.83%.

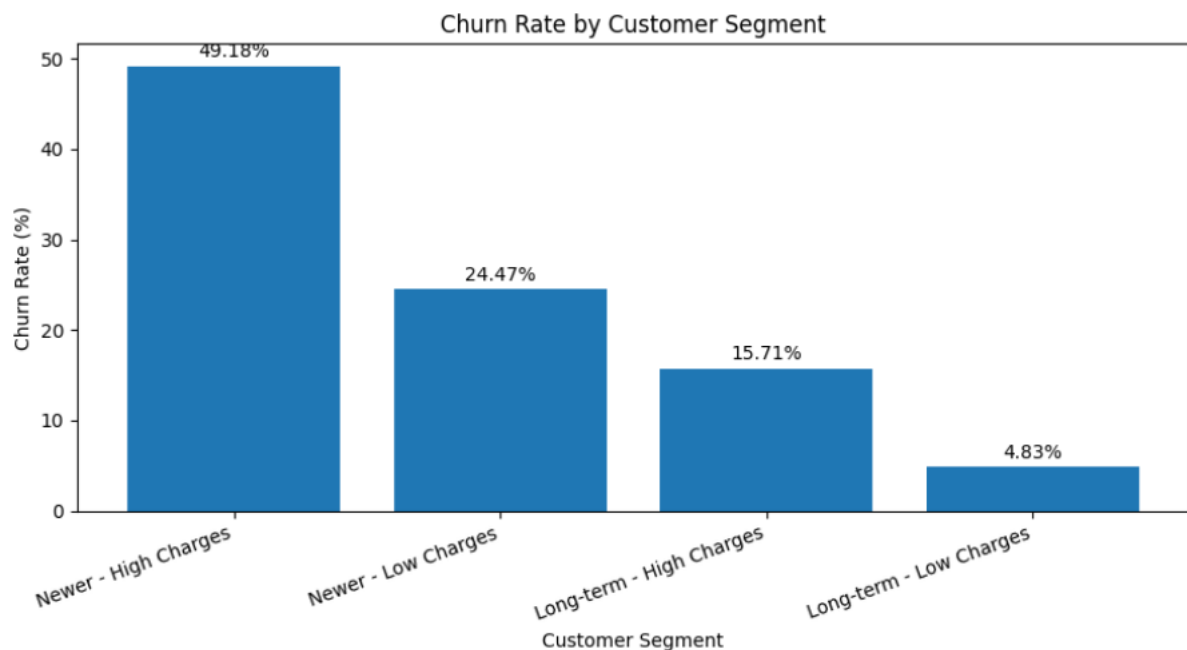


Figure 3: Churn rate by customer segment

This comparison shows that churn was much higher among the newer customer segments, especially newer customers with high monthly charges. Long-term customers had lower churn rates in both charge groups.

6. Key Findings

The clustering analysis identified four different customer segments based on tenure and monthly charges. The results showed clear differences in churn rates between these groups.

The Newer – High Charges segment had the highest churn rate at 49.18%, while the Long-term – Low Charges segment had the lowest churn rate at 4.83%. Another interesting result was that the Long-term – High Charges group had higher average monthly charges (\$93.08) than the Newer – High Charges group (\$81.23), but its churn rate was much lower at 15.71%.

Overall, the results show that churn patterns are different across customer segments based on tenure and monthly charges. These clustering results can be used by the team for further analysis and to support the development of customer retention strategies.